

Theorem (Bernoulli's Inequality). *Let $x \in \mathbb{R}$ satisfy $x \geq -1$ and let $n \in \mathbb{Z}$ satisfy $n \geq 0$. Then*

$$(1+x)^n \geq 1+nx.$$

Proof. **1. Base cases.**

Case $n = 0$.

$$(1+x)^0 = 1 \quad \text{and} \quad 1+0 \cdot x = 1.$$

Hence $(1+x)^0 = 1+0x$, so the inequality holds with equality.

Case $n = 1$.

$$(1+x)^1 = 1+x \quad \text{and} \quad 1+1 \cdot x = 1+x.$$

Again equality holds.

Both base cases are verified directly.

2. Inductive hypothesis. Assume that for some integer $k \geq 0$ the inequality

$$(1+x)^k \geq 1+kx \tag{IH}$$

holds for every real $x \geq -1$.

3. Inductive step. Since $x \geq -1$, we have $1+x \geq 0$. Multiplying both sides of (IH) by the non-negative factor $1+x$ preserves the inequality:

$$(1+x)^k(1+x) \geq (1+kx)(1+x). \tag{1}$$

The right-hand side expands as

$$(1+kx)(1+x) = 1+(k+1)x+kx^2. \tag{2}$$

Thus

$$(1+x)^{k+1} \geq 1+(k+1)x+kx^2. \tag{3}$$

Because $k \geq 0$ and $x^2 \geq 0$, the term kx^2 is non-negative. Dropping this non-negative term yields

$$(1+x)^{k+1} \geq 1+(k+1)x, \tag{1}$$

which is precisely Bernoulli's inequality for the exponent $k+1$.

4. Conclusion by induction. The inequality holds for $n = 0$ and $n = 1$ (the base cases) and the inductive step shows that truth for $n = k$ implies truth for $n = k+1$. By the principle of mathematical induction, the inequality

$$(1+x)^n \geq 1+nx$$

holds for every integer $n \geq 0$ and all real numbers $x \geq -1$.

Equality case. Equality occurs precisely in the following situations:

- $n = 0$, where both sides equal 1;
- $n = 1$, where $1 + x = 1 + x$;
- $x = 0$, for which $(1 + 0)^n = 1 = 1 + n \cdot 0$ for any n .

For all other admissible pairs (x, n) with $x > -1$ and $n \geq 2$, the term kx^2 (or a higher-order binomial term) is positive, so the inequality is strict. $\square$